Research Statement

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Overview

My research interests lie at the interface of number theory, representation theory, and harmonic analysis and are motivated by the Langlands program. My current work focuses on periods and L-functions. In my thesis, I study the realization of L-functions through zeta integrals on Braverman–Kazhdan Schwartz spaces, motivated by the relation between L-functions and periods on spherical varieties within the framework of the relative Langlands program. I also study special L-values and archimedean period relations from arithmetic, algebraic, and analytic perspectives. These interests lead me to local questions in the representation theory of real groups arising from global automorphic problems.

- **Archimedean zeta integrals for split orthogonal Gan–Gross–Prasad pairs.** In my thesis, under the direction of Yiannis Sakellaridis, I develop a uniform geometric approach to archimedean zeta integrals for the split orthogonal Gan–Gross–Prasad pairs. The aim is to recover the predicted half L-factors from integrals on Braverman–Kazhdan Schwartz spaces on $(\mathrm{SO}_n/U_n) \times (\mathrm{SO}_{n+1}/U_{n+1})$, using orbit-adapted blow-ups and the combinatorics of Weyl-group actions; see §1.
- **Arithmetic properties of intertwining operators and rationality of ratios of special L-values.** In joint work with A. Raghuram, I study the arithmetic properties of standard intertwining operators at complex archimedean places and their action on relative Lie algebra cohomology. This project is motivated by the role of these operators in the Eisenstein-cohomological approach to studying the algebraicity of ratios of special L-values, with possible connections to motivic L-functions; see §2.

Contents

1 Archimedean zeta integrals and relative Langlands **2**
2 Arithmetic properties of intertwining operators **5**
3 Future research plan **8**
References **9**

1. Archimedean zeta integrals and relative Langlands

How can period integrals on spherical varieties recover the local L-factors predicted by the relative Langlands program? Sakellaridis [Sak13] reformulates this question for $X = H\backslash G$ in terms of H-invariant, B/U-equivariant distributions on the Braverman–Kazhdan Schwartz space of G/U.

Over p-adic fields, he constructs H-distinguished spherical test vectors in $\mathcal{S}_{\mathrm{BK}}(G/U)$ on which the zeta distribution evaluates to special values of the half L-factor $L_X^{\frac{1}{2}}$ [Sak13, Definition 7.2.3]. My thesis develops the archimedean analogue for orthogonal Gan–Gross–Prasad pairs

$$(G, H) = (\mathrm{SO}_n \times \mathrm{SO}_{n+1}, \mathrm{SO}_n).$$

Rather than constructing a single distinguished test vector, I recover the predicted half L-factor as the greatest common divisor of the family of zeta integrals on the Braverman–Kazhdan space, in the sense made precise below.

The main result

Let $G = \mathrm{SO}_n \times \mathrm{SO}_{n+1}$ and $H = \mathrm{SO}_n$ be the split orthogonal Gan–Gross–Prasad pair over $\mathbb{R}$, with H diagonally embedded. The spherical variety under consideration is $H\backslash G$.

Fix a Borel subgroup $B \subset G$, its unipotent radical U, and the maximal split torus $A = B/U$. The spectral parameters naturally belong to $X^*(A_{H\backslash G}) \otimes_{\mathbb{Z}} \mathbb{C}$, where $A_{H\backslash G}$ is the quotient of A defined in [Sak13, §2.1]. In this case, the two tori agree up to isogeny, so we use A to describe the complexified parameter space.

For $\chi \in \Lambda_{\mathbb{C}}^{(n)} := X^*(A) \otimes_{\mathbb{Z}} \mathbb{C}$, define

$$\Delta_\chi(g) = \int_{\mathcal{O}} g(y)\, \chi'^{-1}(y)\, \mathrm{d}\mu, \qquad g \in \mathcal{S}(G/U).$$

Here $\mathcal{O} \subset G/U$ is the open $H \times A$-orbit, and $\mathrm{d}\mu$ is a fixed A-eigenform. After fixing an orbit representative, we identify χ' with the corresponding eigenfunction on $\mathcal{O}$. The parameter $\chi \in \Lambda_{\mathbb{C}}^{(n)}$ is the inducing character of the principal series $I(\chi)$. Its relation to χ' incorporates the character of the eigenmeasure, following the convention in [Sak13, §3.6].

Recall that the Braverman–Kazhdan space on G/U is the algebraic sum

$$\mathcal{S}_{\mathrm{BK}}(G/U) = \sum_{w \in W} \mathcal{F}_w \mathcal{S}(G/U).$$

Here $\mathcal{S}(G/U)$ is the ordinary Schwartz space, and W is the little Weyl group, which in this case is the Weyl group of G. The Fourier transforms $\mathcal{F}_w$ are defined in [BK99] and recalled in [Sak13, §3.8].

Write

$$L_X^{\frac{1}{2}}(\chi) = \prod_{\check{\theta} \in \Theta^+} \Gamma_{\mathbb{R}}\left(\tfrac{1}{2} + \check{\theta}(\chi)\right), \qquad \Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2), \tag{1}$$

for the half L-factor in this normalization, where Θ^+ is the set of positive virtual weighted colors as defined in [Sak13], which we can also identify with the cocharacters $\theta^\vee$ in $X_*(A)$ whose image is the stabilizer of each orbit descending to A.

Main theorem. *For $2 \le n \le 8$ and every $g \in \mathcal{S}_{\mathrm{BK}}(G/U)$, the zeta integral $\Delta_\chi(g)$ admits meromorphic continuation to $\Lambda_{\mathbb{C}}^{(n)}$. For g fixed, the function $\chi \longmapsto (L_X^{\frac{1}{2}}(\chi))^{-1}\Delta_\chi(g)$ is holomorphic on all of $\Lambda_{\mathbb{C}}^{(n)}$. Moreover, for every $\chi_0 \in \Lambda_{\mathbb{C}}^{(n)}$ there exists $g \in \mathcal{S}_{\mathrm{BK}}(G/U)$ such that*

$$\left(L_X^{\frac{1}{2}}(\chi)\right)^{-1} \Delta_\chi \Big|_{\chi=\chi_0} (g) \neq 0. \tag{2}$$

Equivalently, if $\mathcal{M}$ is the sheaf generated over $\mathcal{O}_{\Lambda_{\mathbb{C}}^{(n)}}$ by meromorphic functions $\chi \mapsto \Delta_\chi(g)$ as g varies over $\mathcal{S}_{\mathrm{BK}}(G/U)$, then $\mathcal{M} = L_X^{\frac{1}{2}} \mathcal{O}_{\Lambda_{\mathbb{C}}^{(n)}}$. This is the precise meaning of the greatest-common-divisor assertion: the normalized integrals are holomorphic and have no common zero.

Idea of proof: geometric and combinatorial approaches

The theorem requires holomorphy of the normalized integrals and nonvanishing of the family at every parameter. Both assertions can be studied using ordinary Schwartz functions on G/U: for $w \in W$ and $g \in \mathcal{S}(G/U)$, the normalized functional equations give

$$\frac{\Delta_\chi(\mathcal{F}_w g)}{L_X^{\frac{1}{2}}(\chi)} = \frac{\Delta_{w\chi}(g)}{L_X^{\frac{1}{2}}(w\chi)}. \tag{3}$$

Thus, moving the test function by a Fourier transform is equivalent to moving the spectral parameter by the Weyl group. Every $f \in \mathcal{S}_{\mathrm{BK}}(G/U)$ is a finite sum $f = \sum_{w\in W} \mathcal{F}_w g_w$ with $g_w \in \mathcal{S}(G/U)$, and hence

$$\frac{\Delta_\chi(f)}{L_X^{\frac{1}{2}}(\chi)} = \sum_{w\in W} \frac{\Delta_{w\chi}(g_w)}{L_X^{\frac{1}{2}}(w\chi)}.$$

Holomorphy for ordinary Schwartz functions throughout $\Lambda_{\mathbb{C}}^{(n)}$ therefore implies holomorphy for the entire Braverman–Kazhdan space. Once holomorphy is known, nonvanishing on the Braverman–Kazhdan space at every parameter $\chi_0 \in \Lambda_{\mathbb{C}}^{(n)}$ is equivalent to the following:

$$\forall \chi_0 \in \Lambda_{\mathbb{C}}^{(n)} \quad \exists w \in W,\ g \in \mathcal{S}(G/U) \quad \text{such that} \quad \left(\mathrm{ev}_{w\chi_0} \circ \frac{\Delta_\chi}{L_X^{\frac{1}{2}}(\chi)}\right)(g) \neq 0. \tag{4}$$

All values are taken after holomorphic continuation. Thus one suitable representative in each Weyl orbit suffices.

Therefore, the problem is divided into two parts.

- *On the geometric side*, I determine the possible poles of $\Delta_\chi(g)$ for $g \in \mathcal{S}(G/U)$ using a tower of B/U-equivariant blow-ups of G/U. The inverse image of the open-orbit boundary has B/U-stable divisorial components. Following the approach to local zeta functions in [Ig00], I study the poles of $\Delta_\chi(g)$ by studying the geometry of the intersections of these divisors.

  The corank-one $H \times B/U$-orbits in G/U are in bijection with the cocharacters $\theta^\vee \in \Theta^+$. By construction, the blow-down image of each B/U-stable divisor is dominated by exactly one corank-one orbit. We therefore denote these divisors by $D_{\theta^\vee}$.

  For $P \subset \Theta^+$, we call the exact stratum of $\bigcap_{\theta^\vee \in P} D_{\theta^\vee}$, obtained by removing all unselected boundary components, the *P-intersection*. We call P *geometrically good* if the P-intersection is

nonempty and locally smooth, the full reduced boundary has simple normal crossings along it, and the restricted blow-down is an isomorphism onto its locally closed image.

Recall that $L_X^{\frac{1}{2}}(\chi)$ is a product of Gamma functions with poles but no zeros. Let $\mathcal{H}^p_{\theta^\vee} = \left\{\chi : \theta^\vee(\chi) = -\frac{1}{2} - 2m, m \in \mathbb{Z}_{\geq 0}\right\}$. The geometry enters through the following result: if P is geometrically good, then for every χ_0 such that

$$\chi_0 \in \left(\bigcap_{\theta^\vee \in P} \mathcal{H}^p_{\theta^\vee}\right) \setminus \left(\bigcup_{\theta^\vee \notin P} \mathcal{H}^p_{\theta^\vee}\right)$$

there exists $g \in \mathcal{S}(G/U)$ such that

$$\lim_{\chi \to \chi_0} \left(\Delta_\chi(g)/L_X^{\frac{1}{2}}(\chi)\right) \neq 0.$$

Thus the quotient $\Delta_\chi(g)/L_X^{\frac{1}{2}}(\chi)$ has a nonzero value at χ_0 after holomorphic continuation, and we say that g supports poles of type P.

I used ChatGPT to investigate geometrically good strata in arbitrary rank n. After I explained the inductive structure and the blow-up construction, it helped me compute the geometrically good sets P in the initial low-rank cases and check proposed higher-rank candidates. These AI-assisted computations of low-rank examples enabled me to conjecture and then prove several generating rules for constructing geometrically good strata inductively.

Weekly usage limits and the time required for these computations made the process more demanding than one might expect. For example, finding a rule that generated the last two missing good strata for $n = 8$ took more than one week. Nevertheless, ChatGPT substantially reduced the time needed to test proposed rules against examples and helped me formulate conjectures.

- *On the spectral side*, I study the Weyl action on the *position* $\mathrm{Pos}(\chi) = (-P, Z)$ of $\chi \in \Lambda_{\mathbb{C}}^{(n)}$. Here P and Z record the pole and zero loci of the Gamma ratios occurring in the Fourier functional equations. More explicitly,

$$P = \{\theta^\vee \in \Theta^+ : \chi \in \mathcal{H}^p_{\theta^\vee}\}, \qquad Z = \{\theta^\vee \in \Theta^+ : \chi \in \mathcal{H}^z_{\theta^\vee}\},$$

where $\mathcal{H}^z_{\theta^\vee} = \left\{\chi : \theta^\vee(\chi) = \frac{1}{2} + 2m, m \in \mathbb{Z}_{\geq 0}\right\}$. Clearly, the sets P and Z are disjoint, and their union is a subset of Θ^+. The Weyl action on χ induces an action on these positions.

The goal is to move every realizable position to an optimal position $(-P', Z')$ for which P' is geometrically good. The local geometric and residue calculations then give nonvanishing at the translated parameter; the normalized functional equations transfer it to the original parameter on the Braverman–Kazhdan space.

For $n \leq 4$, I designed an ordering procedure that moves any parameter to an optimal position by Weyl-group action. For $5 \leq n \leq 8$, I verified that the ordering procedure, followed by additional Weyl-group moves suggested by the geometric rules, moves every realizable position to an optimal one. AI also helped me discover the ordering procedure by quickly finding counterexamples to incorrect versions of the algorithm.

2. Arithmetic properties of intertwining operators

The Eisenstein-cohomological approach to special L-values requires arithmetic information about local standard intertwining operators [HR20, Rag23]. In joint work with A. Raghuram, I extend Raghuram's result [Rag25] for $\mathrm{GL}_N(\mathbb{C})$ to connected complex reductive groups. Under certain criticality hypotheses, I prove that the standard intertwining operator, normalized by the desired ratios of Langlands–Shahidi local L-factors, identifies the prescribed rational lines in bottom-degree cohomology. Together with the p-adic arithmetic results [Rag23], this supplies local input for the Eisenstein-cohomological study of the rationality of ratios of L-values.

The local problem at archimedean places

Sources and targets. Let $\mathrm{G} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}}\,\mathrm{G}_0$, where G_0 is connected and reductive over $\mathbb{C}$, and regard $G = \mathrm{G}(\mathbb{R}) = \mathrm{G}_0(\mathbb{C})$ as a real Lie group. Let $P = L_P N_P$ be a standard maximal parabolic. Write the full inducing datum as $\tau \otimes \underline{w}_P$, an irreducible essentially tempered cohomological representation of L_P, with τ tempered and $\underline{w}_P$ a possible algebraic central character.

Consider the family of normalized parabolic induction, allowing s-parametrized twisting by δ_P^{-s/s_0}, $s \in \mathbb{C}^*$, where δ_P is the modulus character of P:

$$\underline{w}_{P,s} = \underline{w}_P\, \delta_P^{-s/(2s_0)}, \qquad \pi_s = \mathrm{Ind}_P^G(\tau, \underline{w}_{P,s}), \qquad s \in \mathbb{C}.$$

The varying twists are smooth quasi-characters, and we call s_0 *the point of evaluation* [RZ, Definition 3.10] if, at $s = s_0$, the twist cancels the half-modulus in normalized induction so that $\Pi := \pi_{s_0} = {}^a\mathrm{Ind}_P^G(\tau, \underline{w}_P)$ is the algebraic (unnormalized) induction of $\tau \otimes \underline{w}_P$.

We prove that π_s are in fact irreducible principal series representations $I(\chi_s)$. Moreover, let w_P be the Weyl element carrying the opposite of P to its standard associate $Q = L_Q N_Q$. The standard intertwining operator $T_{\mathrm{st}}(s)$ is an isomorphism between $I(\chi_s)$ and $I({}^{w_P}\chi_s)$. We show that, as stated in the combinatorial lemma, at $s = s_0$, the latter is another algebraic induction from the associate parabolic Q of an irreducible essentially tempered cohomological representation:

$$I({}^{w_P}\chi_{s_0}) = {}^a\mathrm{Ind}_Q^G(\tilde{\tau}, \underline{w}_Q) =: \tilde{\Pi}.$$

The two algebraic inductions $\Pi, \tilde{\Pi}$ show up in the infinite part of the global Eisenstein cohomology, with a common algebraic coefficient system $\mathcal{V}_{\underline{\lambda}}$.

Criticality.

The adjoint action of the dual Levi on the Lie algebra of the dual unipotent radical decomposes into graded representations r_j. Their local L-factors give the normalizing factor

$$\prod_j \frac{L_{\mathbb{C}}(js, \tau \otimes \underline{w}_P, r_j)}{L_{\mathbb{C}}(1 + js, \tau \otimes \underline{w}_P, r_j)}, \tag{5}$$

used in the main theorem. Let $\phi_{\tau\otimes\underline{w}_p}$ be the complex Langlands parameter of the full Levi datum, and put $V_j = r_j \circ \phi_{\tau\otimes\underline{w}_p}$. An argument u is critical if it lies on the integer or half-integer lattice determined by the algebraic infinity type of V_j and both $L_{\mathbb{C}}(u, V_j)$ and $L_{\mathbb{C}}(1-u, V_j^\vee)$ are finite. Here $\vee$ denotes the dual Weil representation, not the associate-induction target; see [RZ, Definition 5.7] and [HR20, §7.2.2]. Our hypothesis is

$$\boxed{js_0 \text{ and } 1 + js_0 \text{ are both critical for every } r_j.}$$

Thus (5) at $s = s_0$ is a product of ratios of consecutive critical local values. This is the manuscript's hyper-criticality condition; the local theorem does not assume a global motivic realization.

The combinatorial lemma: a common coefficient system and a common degree. Let $\underline{\mu}$ be the pure weight of L_P attached to $\tau \otimes \underline{w}_P$. Under the consecutive-criticality hypothesis above, I prove a combinatorial lemma for connected complex reductive groups, generalizing [HR20, Lemma 7.14]. It has two consequences.

(1) There are Kostant representatives $\underline{s}$ for P and $\underline{s}' = (w_P, w_P)\underline{s}$ for Q such that $\underline{\lambda} = \underline{s}^{-1} \cdot \underline{\mu}$ is dominant and integral, and $\underline{\mu}' = \underline{s}' \cdot \underline{\lambda}$ is pure for L_Q. Thus $\widetilde{\Pi}$ is realized as algebraic induction from an essentially tempered cohomological representation $\widetilde{\tau} \otimes \underline{w}_Q$ of L_Q.

(2) The representatives have the same absolute Weyl-group length:

$$\ell_{\mathrm{abs}}(\underline{s}) = \ell_{\mathrm{abs}}(\underline{s}') = \tfrac{1}{2} \dim_{\mathbb{R}} N_P = \tfrac{1}{2} \dim_{\mathbb{R}} N_Q. \tag{6}$$

Kostant's theorem places $V_{\underline{\mu}}$ and $V_{\underline{\mu}'}$ in $H^{\ell_{\mathrm{abs}}(\underline{s})}(\mathfrak{n}_P, \mathcal{V}_{\underline{\lambda}})$ and $H^{\ell_{\mathrm{abs}}(\underline{s}')}(\mathfrak{n}_Q, \mathcal{V}_{\underline{\lambda}})$, respectively. Moreover,

$$q_0(L_P) + \ell_{\mathrm{abs}}(\underline{s}) = q_0(L_Q) + \ell_{\mathrm{abs}}(\underline{s}') = q_0(G). \tag{7}$$

The minimal K-types of both representations therefore support relative Lie algebra cohomology in bottom degree $q_0(G)$, with the same coefficient system $\mathcal{V}_{\underline{\lambda}}$.

Main theorem. The strict root inequalities also exclude the Speh–Vogan reducibility obstruction [SV80], proving that Π and $\widetilde{\Pi}$ are irreducible principal series [RZ, §7.3]. Let $\mathfrak{g} = \mathrm{Lie}_{\mathbb{R}}(G) \otimes_{\mathbb{R}} \mathbb{C}$ and $K = C_G Z(G)$, with C_G maximal compact. Write $\Pi \simeq I(\chi)$ and $\widetilde{\Pi} \simeq I(\widetilde{\chi})$, where $\widetilde{\chi} = w_P \chi$ and the fixed Borel characters include the shifts at s_0.

Theorem (Raghuram–Zou). *Under the criticality assumption above, $T_{\mathrm{st}}(s)$ is holomorphic at s_0, and its value $T_{\mathrm{st}}(s_0) : \Pi \xrightarrow{\sim} \widetilde{\Pi}$ is an isomorphism of smooth representations. The independently fixed rational generators described below span the bottom-degree cohomology groups:*

$$H^{q_0(G)}(\mathfrak{g}, K; \Pi \otimes \mathcal{V}_{\underline{\lambda}}) = \mathbb{C}[J_\chi], \quad H^{q_0(G)}(\mathfrak{g}, K; \widetilde{\Pi} \otimes \mathcal{V}_{\underline{\lambda}}) \quad = \mathbb{C}[J_{\widetilde{\chi}}].$$

The induced map satisfies

$$T_{\mathrm{st}}^{\bullet}(s_0)[J_\chi] = a \left(\prod_j \frac{L_{\mathbb{C}}(j s_0, \tau \otimes \underline{w}_P, r_j)}{L_{\mathbb{C}}(1 + j s_0, \tau \otimes \underline{w}_P, r_j)} \right) [J_{\widetilde{\chi}}], \qquad a \in \mathbb{Q}^{\times}. \tag{8}$$

Dividing the induced map by (5) at s_0 therefore identifies the prescribed rational lines, not merely the underlying complex spaces [RZ, Theorem 8.2].

Idea of the proof

Rational structures. A reduced expression $w_P = s_{\alpha_d} \cdots s_{\alpha_1}$ gives intermediate characters $\chi^{(a)} = w_{(a)}\chi$, where $w_{(a)} = s_{\alpha_a} \cdots s_{\alpha_1}$, $\chi^{(0)} = \chi$, and $\chi^{(d)} = \widetilde{\chi}$. Fix ordered Chevalley root vectors, a rational highest-weight model of $\mathcal{V}_{\underline{\lambda}}$, and pinned Weyl representatives. Put $\mathscr{H}_G(\chi) = H^{q_0(G)}(\mathfrak{g}, K; I(\chi) \otimes \mathcal{V}_{\underline{\lambda}})$. Evaluation at 1 and the fixed Kostant coefficient map give

$$\mathscr{K}_\chi^G : \mathscr{H}_G(\chi) \xrightarrow{\sim} \bigotimes_j \mathscr{H}^0(\chi^j), \qquad \mathscr{K}_\chi^G([J_\chi]) = \Omega_\chi. \tag{9}$$

Here $\chi = \boxtimes_j \chi^j$ in fixed torus coordinates, $\mathscr{H}^0(\chi^j) = H^0(\mathfrak{t}_j, U(1); \mathbb{C}_{\chi^j} \otimes \mathbb{C}_{(\chi^j)^{-1}})$, and $\Omega_\chi = \bigotimes_j (1_{\chi^j} \otimes 1_{(\chi^j)^{-1}})$. Inverse Borel evaluation of the specified Kostant cocycle constructs J_χ, whose class spans the rational line $\mathscr{H}_G(\chi)_{\mathbb{Q}} = (\mathscr{K}_\chi^G)^{-1}(\mathbb{Q}\Omega_\chi) = \mathbb{Q}[J_\chi]$. The same construction applies to every $\chi^{(a)}$, fixing all generators independently of T_{st}. Their degree is $q_0(G)$; degree zero refers only to the torus cohomology [RZ, §8.2.2].

The SL_2 comparison. Put $H = \mathrm{SL}_2(\mathbb{C})$, $C_H = \mathrm{SU}(2)$, and $\mathfrak{h} = \mathrm{Lie}_{\mathbb{R}}(H) \otimes_{\mathbb{R}} \mathbb{C}$. At a simple-reflection step $\chi^{(a+1)} = s_\alpha \chi^{(a)}$, the root character is $\psi_\alpha = \chi^{(a)} \circ \alpha^\vee$. The two orientations are

$$\psi_+(z) = z^r \bar{z}^{-t}, \qquad \psi_- = \psi_+^{-1}, \qquad I_\pm = I_H(\psi_\pm), \qquad r, t \in \mathbb{Z}_{>0}.$$

They have the common coefficient system $\mathcal{E}_{r,t} = \mathrm{Sym}^{r-1}(\mathbb{C}^2) \boxtimes \mathrm{Sym}^{t-1}(\mathbb{C}^2)$, with the action on the second factor composed with complex conjugation. The evaluation–Kostant construction fixes generators $[J_\pm]$ of $H^1(\mathfrak{h}, C_H; I_\pm \otimes \mathcal{E}_{r,t})$. I construct cochains $J_\pm^{\mathrm{test}}$ containing the test vectors used to compute the rank-one intertwining integral, and verify that their classes satisfy

$$[J_+^{\mathrm{test}}] = [J_+], \quad [J_-^{\mathrm{test}}] = (-1)^r [J_-].$$

With the fixed root measure $2\,dx\,dy$, the integral calculation gives

$$T_H(\psi_+)^\bullet [J_+] = (-1)^r \frac{2\pi}{r} [J_-], \quad T_H(\psi_-)^\bullet [J_-] = (-1)^t \frac{2\pi}{t} [J_+].$$

The factors $2\pi/r$ and $2\pi/t$ equal $c(\psi_+)$ and $c(\psi_-)$, respectively, where $c(\psi) = L_{\mathbb{C}}(0, \psi)/L_{\mathbb{C}}(1, \psi)$.

Compatibility with induction in stages. The rank-one calculation must now be transferred to the independently chosen rational generators for G. Induction in stages through $P_\alpha = L_\alpha U_\alpha$ first takes outer nilradical cohomology in degree $q_0(G) - 1$. The same Kostant summand occurs on both sides. After cancellation of the remaining central characters, pullback along the pinned root homomorphism $\varphi_\alpha : H \to L_\alpha$ gives comparison maps to the degree-one root cohomology

$$\mathscr{H}_\alpha(\chi^{(a)}) = H^1(\mathfrak{h}, C_H; I_H(\psi_\alpha) \otimes \mathcal{E}_{r,t}).$$

The target is defined in the same way using ψ_α^{-1}. Let T_α^G be the ambient simple-root intertwining operator and $T_H(\psi_\alpha)$ the operator in the rank-one model. I prove that the comparison maps preserve the prescribed rational lines and make the following square commute [RZ, Lemma 8.29]:

$$\begin{array}{ccc} \mathscr{H}_G(\chi^{(a)}) & \xrightarrow{(T_\alpha^G)^\bullet} & \mathscr{H}_G(\chi^{(a+1)}) \\ \Big\downarrow{\scriptstyle S_{\alpha,\chi^{(a)}}} & & \Big\downarrow{\scriptstyle S_{\alpha,\chi^{(a+1)}}} \\ \mathscr{H}_\alpha(\chi^{(a)}) & \xrightarrow[(T_H(\psi_\alpha))^\bullet]{} & \mathscr{H}_\alpha(\chi^{(a+1)}). \end{array} \tag{10}$$

Each $S_{\alpha,\chi^{(a)}}$ sends $[J_{\chi^{(a)}}]$ to a nonzero rational multiple of $[J_{\alpha,\chi^{(a)}}]$, chosen from $[J_\pm]$ according to the orientation.

Consequently, the coefficient of each ambient simple-root operator differs from the computed rank-one coefficient only by the ratio of two nonzero rational comparison constants. Division by the corresponding local L-factor ratio therefore makes its action preserve the ambient rational lines. Composing along the reduced expression of w_P assembles the rootwise L-ratios into (5) at s_0 and proves (8). The irreducibility and nonzero compact-type calculations also establish regularity and invertibility at the evaluation point [RZ, §§8.4–8.6].

3. Future research plan

Archimedean zeta integrals in arbitrary rank and beyond

My next goal is to extend the thesis theorem to arbitrary rank. The blow-up construction applies to general n, and the spectral calculations in the verified cases suggest a uniform approach. Proved geometric rules preserve good strata under specified inheritance operations and produce new strata from lower-dimensional configurations. Some conjectural generating rules remain to be proved, and further rules may be needed.

On the spectral side, the ordering procedure and additional Weyl-group moves suggest an inductive approach. The central problem is to combine the geometric generating rules with a spectral induction showing that every realizable Weyl orbit contains a pole configuration supported by a suitable boundary stratum. Establishing this coverage, together with the required pole bounds and local residue calculations, would extend the holomorphy and nonvanishing theorem to all n. I also expect to discover computable combinatorial descriptions of the underlying geometry and representation theory that support uniform constructions and proofs.

It would also be interesting to investigate archimedean zeta distributions for other spherical varieties beyond the orthogonal GGP pairs and try to recover the half L-factors predicted by Y. Sakellaridis [Sak13]. This would give an archimedean counterpart to his unramified calculations.

Arithmetic at real places, period relations, and nonvanishing

An immediate next step is to generalize Harder–Raghuram's real-place calculation for $\mathrm{GL}_N(\mathbb{R})$ [HR20, Chapter 9] to other real reductive groups. The relevant cohomological representations need not be principal series induced from characters, so the complex-place reduction to torus cohomology cannot simply be repeated. I would like to study the corresponding comparisons for representations involving discrete-series inducing data, keeping track of rational structures and the action of normalized intertwining operators. This would provide further local input for applications of Eisenstein cohomology over totally real fields.

Beyond ratios of critical values, I am also interested in the other problems where the required archimedean input need to be established. For example, the algebraicity results for individual critical L-values worked out in Li–Liu–Sun [LLS24] and Jiang–Liu–Sun–Tian [JLST25] are different but related to the project of A.Raghuram. It would be interesting to investigate their modular-symbols approach in other cases where integral representations of L-functions are known.

All such local results could contribute to global algebraicity and Galois-equivariance statements regrading relevant L-functions, once the necessary global period comparisons are established. I am also interested in how the resulting automorphic period formulas relate to Deligne's conjecture on critical motivic L-values [Del79, DR25].

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